

Surface Hydrides on Fe₂P Electrocatalyst Reduce CO₂ at Low Overpotential: Steering Selectivity to Ethylene GlycolKarin U. D. Calvinho,[⊥] Abdulaziz W. Alherz,[⊥] Kyra M. K. Yap,[⊥] Anders B. Laursen, Shinjae Hwang, Zachary J. L. Bare, Zachary Clifford, Charles B. Musgrave,* and G. Charles Dismukes*Cite This: *J. Am. Chem. Soc.* 2021, 143, 21275–21285

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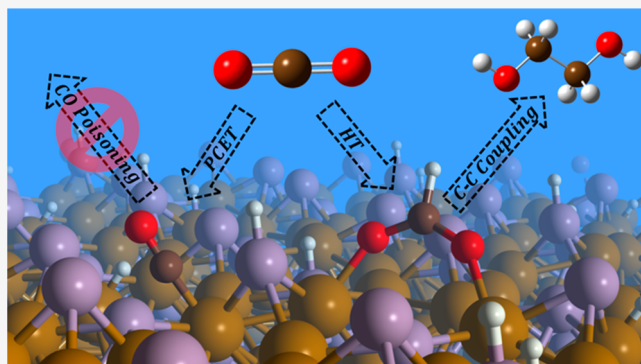


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ABSTRACT: Development of efficient electrocatalysts for the CO₂ reduction reaction (CO₂RR) to multicarbon products has been constrained by high overpotentials and poor selectivity. Here, we introduce iron phosphide (Fe₂P) as an earth-abundant catalyst for the CO₂RR to mainly C₂–C₄ products with a total CO₂RR Faradaic efficiency of 53% at 0 V vs RHE. Carbon product selectivity is tuned in favor of ethylene glycol formation with increasing negative bias at the expense of C₃–C₄ products. Both Grand Canonical-DFT (GC-DFT) calculations and experiments reveal that *formate, not *CO, is the initial intermediate formed from surface phosphino-hydrides and that the latter form ionic hydrides at both surface phosphorus atoms (H@P_s) and P-reconstructed Fe₃ hollow sites (H@P*). Binding of these surface hydrides weakens with negative bias (reactivity increases), which accounts for both the shift to C₂ products over higher C–C coupling products and the increase in the H₂ evolution reaction (HER) rate. GC-DFT predicts that phosphino-hydrides convert *formate to *formaldehyde, the key intermediate for C–C coupling, whereas hydrogen atoms on Fe generate tightly bound *CO via sequential PCET reactions to H₂O. GC-DFT predicts the peak in CO₂RR current density near –0.1 V is due to a local maximum in the binding affinity of *formate and *formaldehyde at this bias, which together with the more labile C₂ product affinity, accounts for the shift to ethylene glycol and away from C₃–C₄ products. Consistent with these predictions, addition of exogenous CO is shown to block all carbon product formation and lower the HER rate. These results demonstrate that the formation of ionic hydrides and their binding affinity, as modulated by the applied potential, controls the carbon product distribution. This knowledge provides new insight into the influence of hydride speciation and applied bias on the chemical reaction mechanism of CO₂RR that is relevant to all transition metal phosphides.



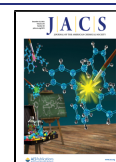
INTRODUCTION

The electrochemical reduction of waste carbon dioxide (CO₂RR) is a promising renewable technology for generating valuable carbon-based products, including monomers for polyethylene and polyethylene terephthalate (PET).^{1,2} Commercializing this technology would aid in the replacement of petrochemicals to tackle global warming. In recent years, there have been many developments in electrocatalysis leading to high activity and selectivity for the conversion of CO₂ to CO,^{3–11} and HCOOH.^{12–20} Recent advances in catalyst doping, promotion, and supports have generated improvements in selectivity to C= products (ethylene^{21–23} and ethanol^{24–26}). Most state-of-the-art catalysts that facilitate C–C coupling are copper-based and produce a mixture of two or more of C₂H₄, C₂H₆, C₂H₅OH, and/or *n*-propanol.^{24,27–38} However, these catalysts suffer from (1) low selectivity and lack of tunability to a single product, (2) high overpotentials, and (3) corrosion instability.

Transition metal phosphides (TMPs) are a new family of CO₂RR electrocatalysts that have disrupted this traditional picture and produced more complex C–C products than observed on other catalysts. Although TMPs were predicted to predominantly catalyze the H₂ evolution reaction (HER) at the expense of CO₂RR activity,³⁹ their high activity in C–C coupling was predicted and demonstrated specifically in the case of nickel phosphides, based on similarity to nickel-group-15 elements present in the active sites of the exergonic class of CO₂ reducing enzymes of acetogenic and methanogenic bacteria.⁴⁰ In particular, while Ni₃P is a poor CO₂RR catalyst and an excellent HER catalyst, both Ni₂P and NiP₂ readily

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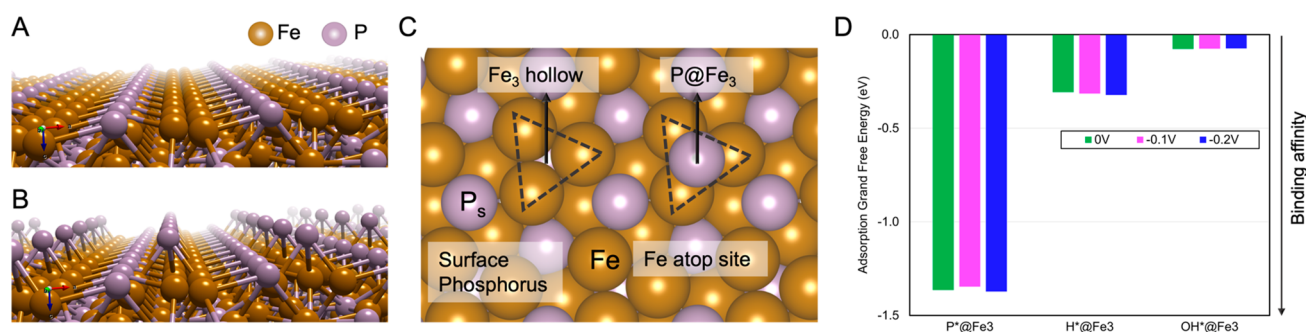


Figure 1. (A) Fe₂P (0001) side view showing the clean surface and (B) reconstructed surfaces. (C) Fe₂P (0001) top view highlighting the Fe₃ hollow site with and without phosphorus reconstruction. (D) Computed adsorption grand free energies (eV) vs applied electrical bias (V vs RHE, pH = 7.5) using GC-DFT. Strong adsorption of P on the Fe site shows that the surface is thermodynamically more stable when P-reconstructed (P*) than not. P* reconstruction also significantly outcompetes H and OH speciation on the Fe₃ site by ~1 eV. Reference states are H₂(aq) for H*, H₂O(aq) for *OH, and bulk white phosphorus for P.

discriminate against H₂ formation. On Ni₂P, CO₂ is reduced to C₄ (2,3-furandiol) with a Faradaic efficiency of 71% at 0 V vs RHE, while, on NiP₂, CO₂ is reduced to C₃ (methylglyoxal) with a Faradaic efficiency of 84% at -0.10 V vs RHE. The discrimination between HER and CO₂RR has been attributed to different chemical mechanisms, with nickel phosphides catalyzing hydride (*H⁻) transfer (HT) to C to produce adsorbed formate bound through its oxygen atom(s) (HCOO⁻*), in contrast to Cu-based catalysts that have been proposed to operate via proton-coupled electron transfer (PCET) to generate carbon monoxide (*CO) as their first intermediate.⁴⁰ Density functional theory (DFT) studies produced key evidence showing that HER on nickel phosphides (e.g., on Ni₂P and Ni₃P)^{41–43} proceeds through reconstructed *P adatoms that bind to native surface Ni₃ “hollows”. These studies further predicted that at increasing negative bias below $U < -0.37$ V, phosphine (PH₃) is released from the Ni₃*P hollow, whereupon two additional H atoms bind to the vacant Ni₃ hollow, generating a Ni–H₂ complex prior to H₂ evolution.

The corresponding di-iron phosphide phase, Fe₂P, is iso-structural with Ni₂P.⁴⁴ Chemical bonding theories predict that iron should stabilize the formation of the P* adlayer more than nickel because elemental Fe atoms have two fewer electrons than Ni, and an ionization potential closer to P. Accordingly, we expect stronger binding of the P adlayer and weaker binding of surface hydrides (Fe₃P–H*) on Fe₂P than on Ni₂P,⁴⁵ with correspondingly higher hydride lability and potentially higher CO₂RR rates. Herein, we examine this hypothesis both experimentally and theoretically, through DFT calculations.

The FeP phase has been previously reported by Sun et al.⁴⁶ to selectively catalyze electrochemical CO₂RR toward methanol and ethanol (Faradaic efficiency of 80.2% for CH₃OH and 14.1% for C₂H₅OH) at -0.2 V vs RHE and low current density. They predicted that CO₂RR proceeds through a proton-mediated mechanism involving a bound *CO intermediate. This pathway is calculated to have a barrier as high as 2.3 eV, which is inconsistent with the low experimentally applied potentials.⁴⁶ In addition to nickel phosphides⁴⁰ and FeP,⁴⁶ other phosphides shown to catalyze CO₂RR are MoP⁴⁷ and Cu₃P⁴⁸ producing formic acid, BP⁴⁹ producing methanol, and CuP₂,⁵⁰ which generates 1-butanol.

Prior studies have observed electrochemical reduction of CO₂ to ethylene glycol (among other products) using Au,⁵¹ Ru,⁵² and Cu^{2,53} as electrocatalysts at potentials more negative

than -0.58 V vs RHE.⁵¹ Herein we report the first observation of CO₂ reduction to ethylene glycol using a transition metal phosphide as the electrocatalyst and achieving a major efficiency improvement by lowering the potential requirement to -0.1 V vs RHE.

In this study, we employ Grand Canonical Electronic Density Functional Theory (GC-DFT) calculations to explore the effects of applied potential on electrocatalytic activity. This state-of-the-art method models the free flow of electrons to and from the electrode to maintain the applied bias by self-consistently solving Schrodinger’s equation for the system with a variable number of electrons that minimizes the grand free energy at a specified electrochemical potential. This enables the prediction of adsorption energies and surface electronic densities as a function of applied potential that traditional DFT approaches cannot access to reveal unanticipated trends.⁵⁴

RESULTS AND DISCUSSION

Catalyst Purity and Active Surface. Phase-pure, polycrystalline iron phosphide (Fe₂P) was synthesized by solid state reaction at 800 °C for 24 h. The high temperature and long reaction time ensured attainment of thermodynamic facet equilibrium. Figure S1 of the Supporting Information (SI) shows that the powder X-ray diffraction pattern (PXRD) of the as-synthesized Fe₂P powder matches all observed peaks of the ICDD reference pattern to confirm that the catalyst was phase-pure within the technique’s detection limit. Scanning electron microscopy (Figure S2) reveals the sample is polydisperse, with particle sizes ranging from 0.5 to 35 μm.

Multiple experimental and theoretical studies have shown that Ni₂P [0001]^{55,56} and therefore, by extension, iso-structural Fe₂P [0001] are the most stable exposed surfaces. Both Ni₂P and Fe₂P [0001] surfaces have repeating layers with M₃P and M₃P₂ stoichiometries, thus averaging to a total stoichiometry of M₂P (See Figure S3). GC-DFT with the CANDLE solvent model predicts that the Fe₃P₂-termination of the Fe₂P [0001] facet is more stable than the Fe₃P termination by approximately 0.34 J/m².

We investigated the surface reconstruction where phosphorus atoms adsorb at Fe₃ hollow sites to form the P ad-layer depicted in Figure 1A–C because there is evidence this minimizes the surface energy of transition metal phosphides.⁵⁷ Figure 1D shows the adsorption energies of H*, OH*, and P* adsorbed at the Fe₃ hollow site on the Fe₃P₂-terminated surface. P* has a considerably more favorable adsorption

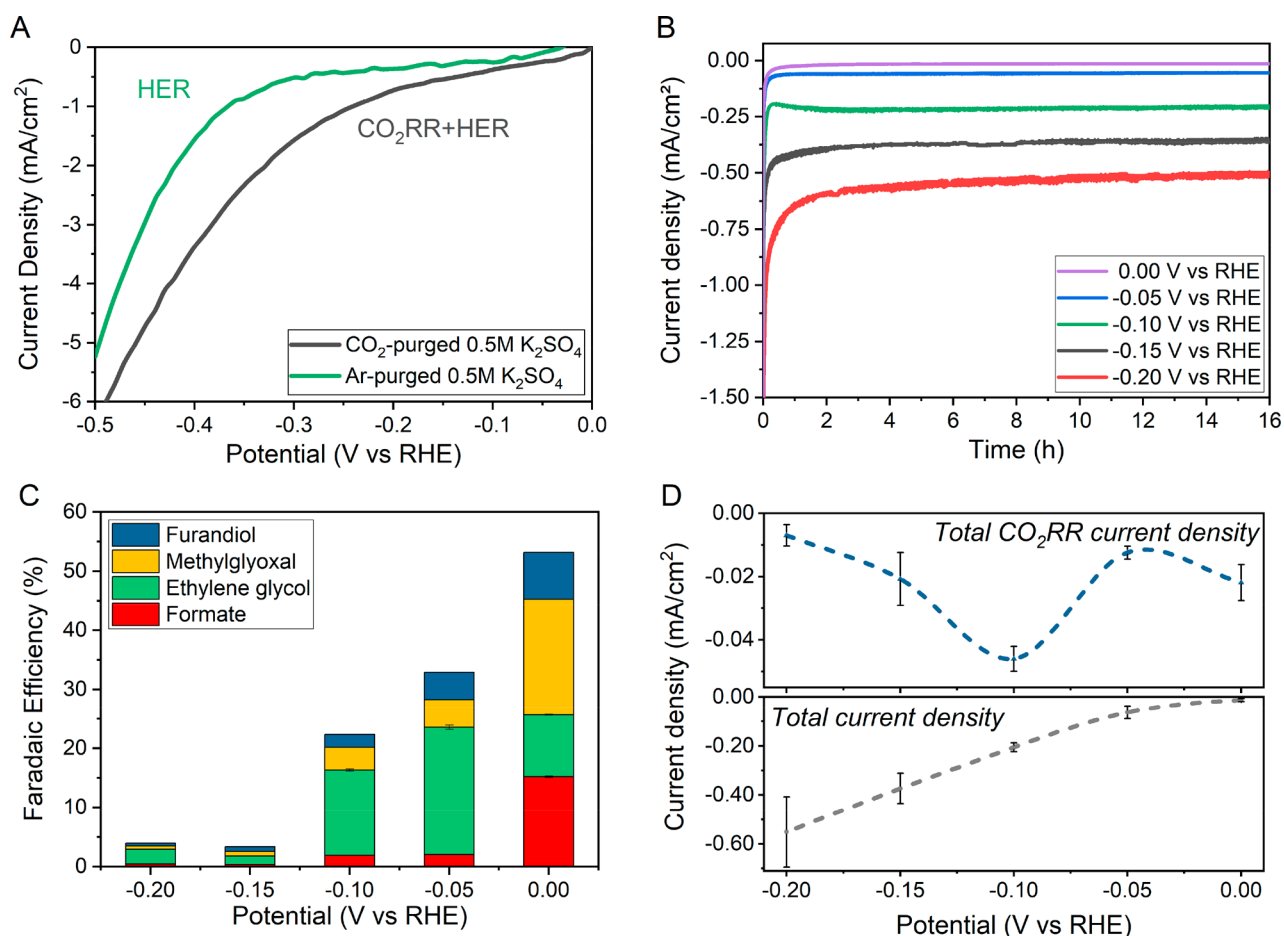


Figure 2. (A) *i*R-corrected linear sweep voltammetry of Fe₂P at 1 mV s⁻¹. Tests were conducted 0.5 M potassium sulfate under argon-purging (green), where the current density is due to HER, and under CO₂ purging (gray), where current is due to CO₂RR and HER. The current density is increased in the presence of CO₂. (B) Chronoamperometry at potentials ranging from 0 to -0.2 V on Fe₂P in CO₂-saturated 0.5 M KHCO₃. (C) Faradaic efficiency for CO₂RR products on an Fe₂P catalyst with a 0.5 M KHCO₃ electrolyte. The remaining Faradaic efficiency is hydrogen. (D) Steady-state total current density in gray and total CO₂ current density in blue (product of the CO₂RR FE and current density).

energy on the Fe₃ hollow site than H* or OH* over the investigated range of applied potentials of 0 V to -0.2 V vs RHE at pH 7.5. Thus, we conclude that the Fe₂P surface is reconstructed with a P* adlayer on Fe₃ hollow sites.

Our GC-DFT calculations indicate that P* adsorption is favored by ~1.4 eV at 100% site coverage; thus, it is likely that the entire Fe₂P [0001] facet reconstructs to form the fully P* covered surface (Figure S2). The predicted Fermi level for the fully phosphorus-reconstructed Fe₃P₂/Fe₂P [0001] surface indicates that the point of zero charge is 0.43 V vs RHE, in agreement with our measured open circuit potential of 0.45 V. Similarly, Rappe et al. found that a P* adatom is energetically favorable on analogous Ni₃ hollow sites of reconstructed Ni₂P and likely serves as the most catalytically active site for HER.^{57–59} Structural reconstruction of the second and third layers was not investigated, as it is assumed that the P* reconstruction is sufficient for describing reaction energetics, as was convincingly demonstrated for the analogous Ni₂P system by Rappe et al.⁵⁷

Surface reconstruction stabilizes Fe₂P by ~0.6 eV/site more than it stabilizes Ni₂P (Figure S4). The much stronger P* affinity on reconstructed Fe₂P is expected based on bonding considerations. Fe⁰ ([Ar]3d⁶ 4s²) has two fewer d-electrons than Ni⁰ ([Ar] 3d⁸ 4s²) and Fe₂P has a smaller energy gap between Fe and P valence orbitals than the gap between the Ni

and P valence orbitals of Ni₂P, which allows for greater hybridization. Indeed, this increase in M–P bond strength for Fe is responsible for the shorter Fe–Fe distance on Fe₂P of ~2.46 Å compared to the ~2.70 Å Ni–Ni distance on Ni₂P.

Electrochemistry. Working electrodes were prepared by mounting an Fe₂P catalyst pellet in a sandwich-type liquid electrolyte cell (as described previously⁴⁰), where the counter and working electrodes are separated by a Nafion 115 membrane. A pressed pellet configuration was chosen to enable reproducible evaluation of the intrinsic activity of the catalyst without variability due to the electrode surface area. However, this configuration has low surface area that requires longer times for product accumulation. Higher CO₂ reduction current densities can be achieved by using gas diffusion electrodes and by nanostructuring the catalyst.

Linear sweep voltammetry was conducted in the presence and absence of CO₂ to determine the CO₂RR+HER and HER activities, respectively. The results are shown in Figure 2A. At all potentials from 0 to -0.50 V, CO₂ activates a higher total current density relative to the Ar-saturated (CO₂-free) condition. Indeed, the CO₂RR+HER contribution to the observed current exceeds the fraction due to HER current alone when positive of -0.40 V, while HER dominates at more negative bias. However, HER increases due to acidification by CO₂ bubbling, and so separation of HER and CO₂RR currents

by this method is not possible. The influence of exogenous carbon monoxide (CO) addition is revealing, as described later.

Representative chronoamperometry traces from 16 h experiments in 0.5 M KHCO₃ at a constant CO₂ flow of 5 sccm are shown in Figure 2B. The porosity of the pressed-pellet catalyst causes an initial induction period of up to 2 h during which the oxidized Fe₂P catalyst surface reduces and equilibrates with adsorbed intermediates concurrently with activation of CO₂RR current.³⁰ A steady-state current is reached thereafter and maintained for the remainder of the experiment. Repeating the chronoamperometry after electrolyte replacement and at any of the selected potentials with the same catalyst pellet reproduces the induction period and steady-state current. This indicates that the induction period is caused by a reversible phenomenon consistent with chemical equilibration of a porous surface with reaction intermediates under applied potential.

CO₂ Reduction Reaction Products. After application of a constant reduction potential for 16 h, the liquid products were analyzed by HPLC and ¹H NMR, while gaseous products were measured by HPLC-GC using TID and RI detection.⁴⁰ Fe₂P catalyzed the reduction of CO₂ and water to five products: formate (C₁), ethylene glycol (C₂), methylglyoxal (C₃), 2,3-furandiol (C₄), and hydrogen. All of these products—except for ethylene glycol—have been reported as CO₂RR products on nickel phosphide catalysts.⁴⁰ The Faradaic efficiency for total CO₂RR and for HER as a function of the applied potential are shown in Figure 2C and Figure S6, respectively. The total CO₂RR Faradaic efficiency (53%) peaks at 0.00 V and decreases with more negative biases: 33% at −0.05 V, 22% at −0.10 V, 3% at −0.15 V, and 4% at −0.20 V. Fe₂P catalyzes production of significantly more ethylene glycol (18% of CO₂RR) at potentials as low as 0.00 V vs RHE (Figure 2C). The ratio of ethylene glycol to other CO₂RR products increases to 65% at both −0.05 V and −0.10 V, although the partial currents shift to favor HER. However, we note that, in terms of electron yield, CO₂R to ethylene glycol requires 10 e[−] vs 2 e[−] to reduce H₂.

Figure 2D compares the total CO₂RR current density from all carbon products to the total current density (HER + CO₂RR) as a function of potential. As expected, the total current density obeys the Butler–Volmer equation with exponentially increasing current density vs applied potential. In contrast, the total CO₂RR current density increased between 0.00 V and −0.10 V to a maximum of 46 μA/cm², then decreases significantly to a local minimum at −0.15 V (12 μA/cm²), followed by a slight increase at −0.20 V at the end of the range. As shown below, this response agrees with our theoretical predictions of the adsorption energies of intermediates as a function of potential. The general trend favoring CO₂RR over HER at low overpotentials suggests thermodynamic control. At increased negative bias, HER outcompetes CO₂RR due to weaker H* binding, fewer steps to the product, and phase separation of gaseous H₂, analogous to the nickel phosphides.⁴⁰ The C_n product selectivity also changes predictably with bias, as described in a later section.

Our GC-DFT calculations show that H* adsorption is possible on three unique sites on the reconstructed Fe₂P surface, which have different electronic properties (Figure 3). We find that at −0.1 V vs RHE, H* adsorption on the P* site is most endergonic (+0.33 eV), followed by the Fe site (+0.07 eV) and least endergonic on the surface P_s site (+0.01 eV).

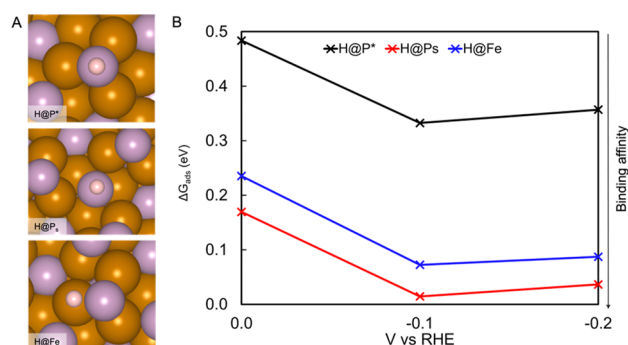


Figure 3. (A) Top-down view of H adsorption on the reconstructed P*, surface P_s, and Fe sites. (B) Adsorption grand free energies of H on the sites depicted in (A) as a function of bias. A minimum in adsorption energy (maximum in binding affinity) is observed for all three sites at −0.1 V.

Although the adsorption grand free energies are all endergonic, they are all readily thermally accessible at 300 K to facilitate CO₂RR. OH* adsorption was also investigated and found to bind less strongly to these sites compared to H*, as shown in Figure S17. The oxidation states and atomic charges obtained using Bader charge analysis are reported in Table 1. These

Table 1. Oxidation States of Adsorbed H Atoms at Three H Adsorption Sites Based on Bader Charge Analysis: Surface Phosphorus Sites (H@P_s), Reconstructed Phosphorus Sites (P*), and Surface Fe Sites (H@Fe)^a

Oxidation States	0 V	−0.1 V	−0.2 V
H@P _s	−1.093 (−0.10)	−1.088 (−0.10)	−1.085 (−0.11)
H@P*	−1.179 (−0.27)	−1.177 (−0.28)	−1.180 (−0.30)
H@Fe	−0.220 (−0.02)	−0.221 (−0.02)	−0.225 (−0.02)

^aThe sum of atomic charges of H and the atom to which it is adsorbed is indicated in parentheses.

indicate that adsorbed H on both P sites, and especially the P* adatom, are hydridic with a charge of −1 (*H[−]), whereas H adsorbed on the Fe atop site has lower electron density and is nearly charge neutral. This suggests that adsorbed hydridic H* on P sites are more likely to carry out concerted two-electron hydride transfer (HT) reactions, whereas the H@Fe sites are more likely to perform one-electron hydrogen atom transfer (HAT) or PCET steps.

Figure 3B reveals that a minimum in the calculated H* adsorption free energies (maximum in the binding affinity) for all surface sites as a function of bias between 0 to −0.2 V occurs at −0.1 V vs RHE. This feature correlates well with the peak in CO₂RR current density measured at −0.1 V (Figure 2D). Taken together, this suggests that the stronger binding of hydrides at −0.1 V and their consequently longer lifetime enable the slower CO₂RR steps to compete better with the faster HER steps.

We note that the adsorption energies are referenced to H₂(aq) and so are offset relative to the solvated proton reference. As the latter energy depends on the local pH of the system at the electrode, which is unknown, we adopted the former reference. Adsorption energies referenced to dissolved

H₂ may be directly correlated to HER activity. For further discussion, see the SI.

Nonlinear electrochemical behavior, such as the changes in H* adsorption energies with applied potential shown in Figures 1D and 3B, arise from a system's intricate electronic structure and its complex dependence on bias, as shown by the projected Density of States (pDOS) plots in Figures S8–S10. As the Fermi level rises with more reducing applied potentials, population of these initially unoccupied states can result in orbital rehybridization and changes in the Helmholtz free energy, as seen in the shift and mixing of states of the pDOS near the Fermi level for the bare reconstructed surface (P*, Figure S8) but not for H adsorbed at this site (H@P*, Figure S9). Consequently, the bias changes the H* well depth and position on the grand free potential energy surface, which changes the H* binding affinity and causes the adsorbate–surface geometries to shift in response.⁵⁴ This complex interplay between surface and adsorbate states, as modified by the applied bias, is essential to understanding the features in Figures 1D and 3B and is uniquely predicted by the GC-DFT approach.

Effect of Applied Bias on CO₂RR and HER Selectivity.

Table 2 shows the equilibrium (thermodynamic) reduction

Table 2. Standard Electrochemical Potentials at pH 7.0 of CO₂RR Half-Reaction Products

Product	Half-Reaction	E^0 (V vs RHE)
Hydrogen	$2(e^- + H^+) \rightleftharpoons H_2$	0.00
Formic acid	$CO_2 + 2(e^- + H^+) \rightleftharpoons HCOOH$	−0.02
Ethylene glycol	$2CO_2 + 10(e^- + H^+) \rightleftharpoons C_2H_6O_2 + 2H_2O$	+0.20
Methylglyoxal	$3CO_2 + 12(e^- + H^+) \rightleftharpoons C_3H_4O_2 + 4H_2O$	+0.02
2,3-furandiol	$4CO_2 + 14(e^- + H^+) \rightleftharpoons C_4H_4O_3 + 5H_2O$	+0.01

potentials (E^0 at pH 7.0 vs RHE) and the number of electrons required to reduce CO₂ to various products taken from literature data,^{60,61} or calculated by Mavrounietis's method of individual group contributions.^{62,63} This shows that ethylene glycol is predicted to be the most thermodynamically favored product by at least 0.18 V over all other products, including

H₂. Thus, we expect ethylene glycol to form unless the reactions are affected by kinetic factors such as high activation barriers or limited reactant availability. The rate of formation and dissociation of intermediates from the electrode surface and their surface mobility must be considered as entropy disfavors forming products with greater numbers of carbons.

For example, the dependence of product yields on applied potential in Figure 2C shows that at all potentials below 0 V, the C₁, C₃, and C₄ products are lost to formation of additional C₂ (ethylene glycol). Because two *C₁ intermediates (formic acid or formaldehyde) plus 2 equiv of *H are required to form ethylene glycol, it is apparent that the *C₂ precursor to produce ethylene glycol is also the precursor to the C₃ and C₄ products. The *C₂ aldol precursor (glycolaldehyde) simply reacts faster with *H to produce more ethylene glycol as the bias decreases than it reacts with the decreasing amount of *C₁ via a slower, entropically disfavored step (Figure 4). We also note that the measured free energy change for the sequential aldol coupling reactions that convert C₁ to C₂ and C₂ to C₃ aldols has a smaller driving force (~6 kJ/mol) for C₂ to C₃.⁴⁰ Thus, both entropic and possibly enthalpic terms disfavor conversion of *C₂ to the *C₃ intermediate. This simple argument explains the change in relative product yields with applied potential on Fe₂P, while operating via the same basic mechanism that is experimentally validated for Ni₂P.⁴⁰

Figure 4 shows the proposed reaction mechanism based on our experiments and calculations, while Table S3 summarizes their equilibrium potentials from literature sources. Figure S7 details all the steps we considered and discuss next. To form ethylene glycol, methylglyoxal, and 2,3-furandiol, the reaction must undergo carbon–carbon coupling through formaldehyde, which was first proposed and experimentally verified in our previous work on Ni₂P⁴⁰ and confirmed by DFT calculations by Rappe et al.⁶⁴ Our GC-DFT calculations reveal that Fe₂P can catalyze CO₂RR via two possible pathways to form formaldehyde, but with very different energetics for the intermediates, as summarized in Figure 5.

Formate Pathway. The first and most likely reaction pathway based on thermodynamics proceeds by the formation of surface bound formate, shown as the pink trace in Figure 5. We identify two routes by which formate can form and bind on the surface. The first route involves a direct HT from a surface

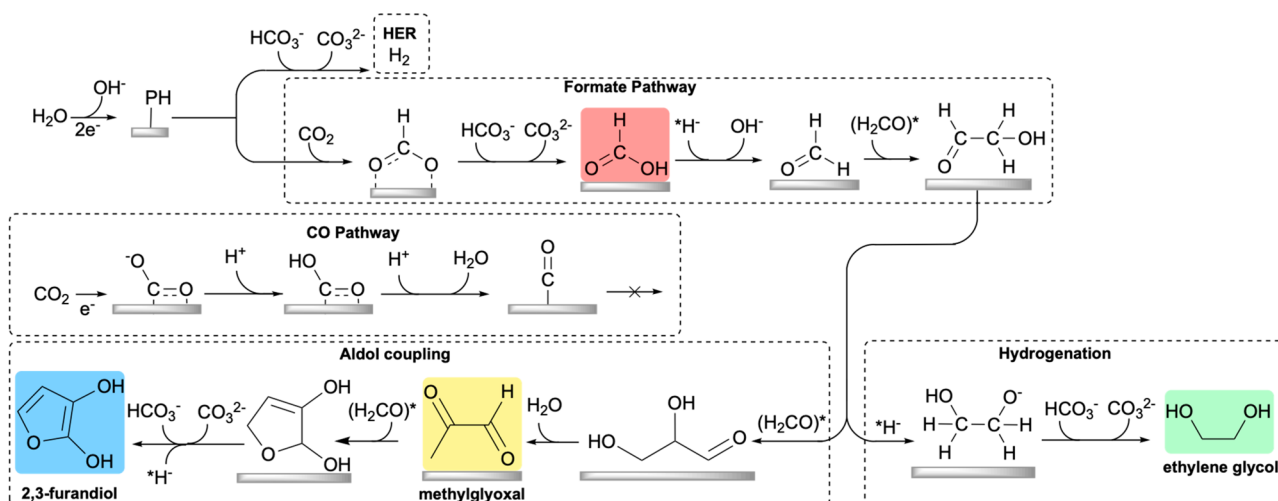


Figure 4. Proposed mechanism for formation of CO₂RR products on Fe₂P.

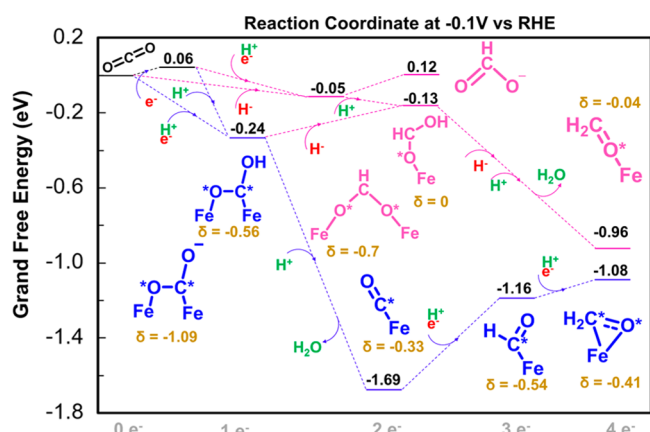


Figure 5. CO₂R Reaction Coordinate Diagram via two pathways: phosphino-hydrides HT (pink) and PCET from Fe sites (blue). Grand free energies relative to aqueous CO₂ are computed using GC-DFT methods described in the SI. The sum of charge densities on all adsorbed atoms, excluding the Fe sites, are shown in gold. The HT pathway involves a surface-bound *formate intermediate, which can then desorb to yield the experimentally observed formate product or undergo further reduction to *formaldehyde. The PCET pathway goes through *COOH and *CO intermediates, followed by two more sequentially uphill PCET steps to arrive at surface-bound *formaldehyde. *CO is unusually stable and is expected to poison the surface and block or slow C product formation at operating potentials. *Formaldehyde is then assumed to undergo C–C coupling to glycolaldehyde by analogy to the experimentally verified pathway on nickel phosphides, followed by hydride transfer to ethylene glycol, as summarized in Figure 4. CO_{2(aq)} and H_{2(aq)} are used as the references for all energies computed in this figure.

hydride on the P_s or P* sites to the carbon of CO₂ to form formate aided by its coordination to a surface Fe site. Although the P_s site is more likely to be occupied by hydrides, the hydrides on the P* site are more reactive owing to their slightly higher electron density (hydricity). The Fe site, however, binds neutral H atoms (Table 1) that are less reactive in reducing aqueous CO₂ to formate. However, they can contribute to forming *formate via another route—HAT to *CO₂^{•−}, the surface bound anion radical. This reaction is restricted to the Fe site, in contrast to the corresponding PCET in which an electron from the surface plus a solvated proton react independently to form *formate. However, the latter PCET

pathway competes with the lower energy pathway that forms *CO and water via the outward-facing oxygen atom of *CO₂^{•−}, resulting in *COOH (Figure 5). GC-DFT predicts that both of these routes to formate formation are thermodynamically feasible, although transition state calculations are necessary to predict which route proceeds at faster rates.

At −0.1 V bias, formate binds to Fe via a Fe–O bond or by bridging two Fe atoms, with the bidentate geometry being energetically favored by −0.05 eV relative to CO₂ (Figure 5). Figure S18 shows that formate production becomes exergonic as the potential is swept from 0 to −0.1 V vs RHE, while the monodentate Fe binding geometry for formate production becomes exergonic at −0.2 V vs RHE. The release of bidentate formate into solution has an energy cost of 0.17 eV at −0.1 V vs RHE, which is thermodynamically accessible at 300 K and explains why formate is observed as a product, albeit at low Faradaic efficiency.

Formate is more likely to undergo protonation to form surface-bound formic acid rather than desorbing. Protonation of bidentate *formate breaks one of its two Fe–O bonds and is exergonic by 0.08 eV at −0.1 V applied bias. HT to the carbon of *HCOOH, followed by protonation of the OH group. The resulting *H₂CO–OH₂ releases a water molecule forming *formaldehyde. This reaction step is highly exergonic by 0.83–0.96 eV, depending on the geometry of *H₂CO, which can readily undergo carbon–carbon coupling with other formaldehyde molecules on nearby Fe sites to form C₂ and higher products, as illustrated in Figure 4. The carbon–carbon coupling step was previously shown to be energetically favorable on Ni₂P,⁶⁴ although the calculations were performed using the computational hydrogen electrode (CHE) model and require further validation, for example by GC-DFT.

CO Pathway. Formaldehyde may potentially form via a *CO intermediate formed by sequential or coupled transfers of electrons and protons to CO₂, as shown by the blue trace in Figure 5. For this to occur, the surface must first bind CO₂ by transferring electron density to it to produce the anion radical *CO₂^{•−} with a net charge of −1.09, as calculated by Bader charge analysis. The resulting *CO₂^{•−} bridges between two Fe atoms in an Fe–C–O–Fe configuration, as illustrated in Figures 4 and 5. At −0.1 V bias, this step is endergonic by +0.06 eV and is even more endergonic at 0 and −0.2 V. Figure S19 shows that while CO₂ binding is endergonic and requires

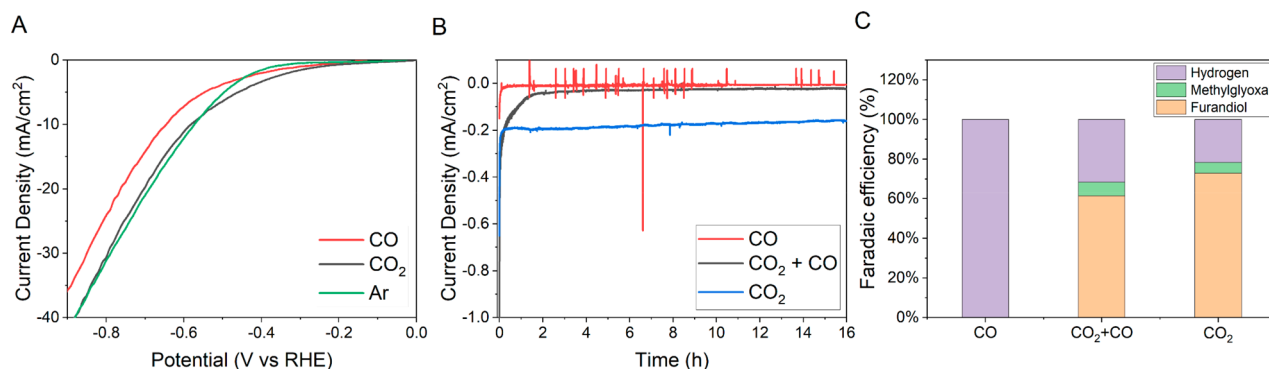


Figure 6. Carbon monoxide slow HER and CO₂RR experiments performed in 0.5 M K₂SO₄ under different purge gases. (A) Linear sweep voltammetry reveals that carbon monoxide reduces the total HER (Ar) and CO₂RR currents (CO₂). (B) Chronoamperometry at −0.1 V vs RHE for 16 h under CO, 50% CO + 50% CO₂, and CO₂, showing that current is significantly reduced in the presence of total or partial CO atmosphere. Spikes are due to noise from H₂ bubble formation. (C) Faradaic efficiency corresponding to B, showing that CO is not converted to carbon products at −0.1 V; while current is suppressed with partial CO atmosphere, there is minimal change in C product distribution.

bending of the highly stable linear CO₂ molecule, it remains kinetically accessible as our upper limit approximation of the transition state energy is computed to be ~0.5 eV. *CO₂[−] is then protonated to form *COOH with no significant change in the geometry as the Fe–C–O–Fe configuration remains intact. The formation of the *COOH intermediate is exergonic with a grand free energy of −0.24 eV relative to aqueous CO₂.

Because surface hydrides are readily available on nearby P_s and P* sites, *COOH could undergo an HT to form surface bound formic acid (*OCHOH) and cross over to the hydride mediated pathway described above, requiring a small energy penalty of 0.11 eV. Otherwise, *COOH can undergo another protonation and a dehydration reaction to form *CO adsorbed to an Fe site. This *CO binds strongly to the Fe site with a computed adsorption grand free energy of −1.69 eV relative to CO₂. Further conversion is suppressed by this deep binding well. Two sequential PCETs or HAT steps are necessary to form *formaldehyde from *CO via an intermediary Fe-bound *CHO. Both of those steps are sequentially uphill for a net free energy cost of +0.61 eV. This high energy cost should introduce a significant kinetic barrier, resulting in poisoning of the surface by *CO, where the Fe sites that catalyze CO₂R to formaldehyde via the formate pathway would be blocked by addition of exogenous CO. Next, we tested this prediction.

In agreement with the theoretical prediction, our experiments depicted in Figure 6 show that, under a CO atmosphere, there is no CO₂RR current observable, while the HER current is significantly suppressed. Under an atmosphere of CO and a constant applied potential of −0.1 V for 16 h, no carbon products were detected by NMR or HPLC, while hydrogen was the only detected product. At a constant applied potential of −0.1 V for 16 h, the current density is suppressed significantly under a gas mixture containing 50% CO₂ and 50% CO. The CO gas mixture partially inhibits the current, lowering the CO₂RR accumulated current (product) cumulatively by 35% relative to pure CO₂ purging, while slightly decreasing the CO₂RR Faradaic efficiency relative to HER. With a mixed CO and CO₂ atmosphere, CO produced a minor or no change in C product selectivity, indicating it neither contributes to C product formation nor modifies the CO₂-dependent pathway to products. These results show that CO is an inhibitor and that the pathway depicted in Figure 5 is a dead-end, consistent with the GC-DFT results showing high barriers to products.

Origin of the C_n Product Selectivity. The ~30% increase in the adsorption energy of bidentate formate at −0.10 V predicts that it has a longer lifetime on the catalyst surface, increasing the likelihood of a second HT to *formaldehyde, which is the obligate precursor to C–C coupling via the aldol mechanism and analogous to the mechanism on Ni₂P.⁴⁰ Indeed, the results shown in Figure 2 predict that the C₂ product dominates relative to all other observed carbon products (C₁, C₃, and C₄) at biases more negative than −0.05 V. This differs from the nickel phosphide catalysts where C₃ and C₄ products are favored at all potentials and C₂ products were not observed at all.⁴⁰ One explanation for this difference is provided by the branching reactions that separate the C₂ and C₃ pathways of the proposed mechanism (Figure 4). The C₂ branch point requires a third *hydride addition, while the C₃ branch point requires a third *formaldehyde addition. The results reported in Figure 3B show that surface hydrides bind weakly and are more labile as the bias increases from −0.1 V to −0.2 V. The resulting increase in hydride reactivity together

with the longer lifetime of adsorption of *formate favors the C₂ pathway relative to C₃ and C₄, consistent with our proposed mechanism.

Catalyst Corrosion Stability. To investigate catalyst stability, inductively coupled plasma emission spectroscopy (ICP) was conducted on the electrolyte after the 16 h catalyst activity test. These results, summarized in Table S8 and Figure S13, indicate that less than 0.2% of Fe and less than 1.5% of P consistently leached from the initially prepared electrode. This behavior is analogous to that of Ni₃P, Ni₂P, and Ni₅P₄ and is postulated to be due to the dissolution of a surface phosphor-oxide layer (formed by air exposure) after which the catalyst is stable.^{58,65} This explanation is corroborated by the XPS data (Figure S14). Based on this, it is expected that the Fe and P dissolution do not indicate an inherent instability of the catalyst under working conditions. In contrast, in this study we observed that the dissolved Fe concentration depends on the applied potential, with stronger reducing potentials resulting in increased Fe dissolution. This result contradicts the thermodynamic behavior of Fe predicted by the Pourbaix diagram at pH 7.5.⁶⁶ However, at more reducing potentials, the gas-evolving HER increases exponentially causing bubble formation that could lead to Fe₂P particles dislodging from the pellet (bubbles in Figure 6). Such dislodged particles would be free to spontaneously oxidize after the cathodic bias is removed, giving rise to an increase in Fe and P concentration in solution at strong bias.

CONCLUSIONS

In this study we show that Fe₂P efficiently converts CO₂ primarily to ethylene glycol and to smaller amounts of formate, methylglyoxal, and 2,3-furandiol at potentials as low as 0 V vs RHE. This compares to the equally efficient CO₂RR on Ni₂P and NiP₂ which produces predominantly C₃ and C₄ products and no C₂ products.⁴⁰ GC-DFT provides an explanation for this shift.

GC-DFT reveals three features critical to CO₂RR catalysis on transition metal phosphides: (1) formation of more ionic hydrides as phosphino-hydrides that are distinctly more reactive, (2) lower free energies of hydride dissociation, and (3) higher C₁ surface affinities (*formate, *formaldehyde, and *CO) relative to C₂ and larger carbon intermediates. Consideration of these features enabled development of a mechanism that accounts for the H/C selectivity and the branching between C₂ and C₃/C₄ pathways at various applied biases. Such insight would not have been possible with the use of conventional DFT calculations and the CHE model, as it fails to consider orbital rehybridization under applied bias. Relative to the iso-structural Ni₂P catalyst, the GC-DFT computed adsorption energies of hydrides on Fe₂P are more weakly bound (more labile) owing to stronger Fe–P bonding. This result is consistent with the preferential formation of shorter C₁ and C₂ products on Fe₂P, in contrast with the longer chain C₃ and C₄ products on Ni₂P, due to the rapid termination of chain growth by hydrogenation of glycolaldehyde (the C₂ intermediate required to form C₃ and C₄ products in the mechanism). A greater hydride lability on Fe₂P also explains why the kinetically favored HER is substantially preferred on this catalyst at comparable potentials.

Through predictions from computational modeling and experimental confirmation, we show that, in contrast to copper-based catalysts, CO is not a reaction intermediate for

C_n products on Fe_2P , but instead poisons the catalyst surface. Strategies for reducing CO poisoning of transition metal phosphides may increase the intrinsic CO_2RR activity.

These results underscore the importance of surface reconstruction on both nickel and iron phosphides and the role of hydride transfer chemistry in CO_2RR that may be more generally applicable to designing new pnictide catalysts. This insight is fundamental to the discovery of new materials for the conversion of CO_2 into organic chemicals.

METHODS

Catalyst Synthesis. Iron metal powder (Fisher Scientific) was annealed under 1% hydrogen (balance Ar) flow for 4 h at 450 °C to remove surface oxides. The reduced iron powder was mixed with a 1.5% molar excess of red phosphorus (Alfa Aesar, 98.9%, 100 mesh), for a total of 10 g of sample per batch. The powder was ground with an agate mortar and pestle for 20 min, transferred to a quartz tube, and then flushed with argon and evacuated to less than 100 mTorr three times. The quartz tubes were sealed and heated at a rate of 0.5 °C min⁻¹ stepwise to 250 °C, 450 °C, 550 °C, and finally to 800 °C. The temperature was maintained for 6 h at each intermediate step to avoid hotspot formation due to the exothermic reaction, and 24 h at the final temperature. The sample was left to cool naturally in the furnace to 60 °C. Powder X-ray diffraction (PXRD) was used to determine what/if additional amounts of phosphorus or iron were needed; these were added, and the synthesis repeated until the catalyst was phase-pure. After synthesis, the Fe_2P powder was washed in 3% HCl, stirring under argon for 2 h, to remove any soluble phosphates. The catalyst was then rinsed and centrifuged five times with water, and then twice with ethanol, followed by drying at room temperature under vacuum.

Electrochemistry. All potentials for experiments performed in this work are referenced to RHE at the pH of 7.5 used throughout this paper. Electrochemical measurements were performed in a custom-made glass-reinforced nylon-6,6 cell, with silicon O-rings and PEEK fittings (IDEX-HS) as previously described.⁴⁰ The working and counter electrodes were separated by a Nafion 115 membrane (Fuel Cell Store). Platinum black deposited on Pt foil (Alfa Aesar, 99.9%) was the counter electrode, and an ALS Hg/ H_2SO_4 was used as the reference electrode, which was calibrated before each experiment against a pristine Ohaus SCE electrode. This SCE was periodically calibrated against a freshly flame-annealed Pt electrode in 0.5 M H_2SO_4 under 1 atm of H_2 . The working electrode was prepared by mixing 1.400 g of the catalyst with 10% (w/w) neutralized Nafion suspension supported on an Al-mesh backing (20 × 20 mesh size, McMaster-Carr) and pressed at 22 ton onto an aluminum die (ϕ = 2 cm). The die with the pressed catalyst pellet served as the working electrode and current collector, with only the Fe_2P surface exposed to the electrolyte (Figure S11). CO_2 (Air Gas, instrument grade, with a Supelco hydrocarbon trap) was supplied through the bottom of the cell to both the working and counter electrodes by a fritted gas dispersion tube (Ace Glass, 5–8 μ m pore size) at a flow rate of 5 sccm (certified MKS P4B mass flow controllers). The electrolysis was run for 16 h at a time with the headspace of the working electrode compartment sampled every hour by gas chromatography. Liquid samples were collected at the end of the electrolysis (see further detail below). Each CO_2RR FE value reflects the average of at least 3 replicates.

Electrochemical measurements were performed with a Gamry 5000E potentiostat. Before each electrolysis experiment, the electrolyte (0.5 M $KHCO_3$, Sigma-Aldrich ACS reagent grade, Chelex-treated⁴¹) was presaturated with CO_2 for an hour. Then, a gas chromatogram was taken to ensure that no air was present in the headspace. A potentiostatic electrochemical impedance spectrum at 0 V vs RHE from 1 Hz to 1 MHz was run to determine the uncompensated resistance (typically between 6 and 10 ohms; see Figure S15). Chronoamperometry was then performed for 16 h with manual IR compensation. Between experiments, the electrochemical

cell was rinsed with Millipore water, and the working electrode catalyst pellet was soaked in Millipore water for 10 min to prevent carryover of product between experiments, and then dried under a vacuum for 10 min to avoid oxidation of the catalyst surface. The catalyst pellet was reused for multiple experiments at varying potentials. By doing this, the longevity of the electrodes, particularly the working electrode, was ensured with no significant difference in product distribution observed as the electrode was reused. Additional experimental replicates using freshly made catalysts were run at all potentials to ensure that the product distribution was consistent across the investigated potential range. The electrochemical surface area was determined as described in the SI.

Gas Chromatography. Detection and quantification of possible headspace products (hydrogen, carbon monoxide, carbon dioxide, methane, ethane, and ethylene) was performed by an autosampling online HP 5890 Series II GC with a 500 μ L sample loop. The GC was fitted with a 6' packed HayeSep D, and a 6' packed MoleSieve 13X column, with thermal conductivity and flame ionization detectors connected in series. Samples were taken before chronoamperometry to ensure that the cell was CO_2 saturated and had no air leaks, and then every hour once chronoamperometry was started. Calibration curves were constructed from certified gas standards (Gasco) by CO_2 dilution using mass flow controllers. The hydrogen calibration was done with *in situ* generated gas through electrolysis of water on platinum, under argon (supplied by an MFC), and diluted post-reaction with CO_2 .

High-Performance Liquid Chromatography (UV/RID). Liquid products were identified and quantified by a PerkinElmer Flexar HPLC equipped with an autosampler, refractive index (RID), and UV–vis detector. An HPX-87H Aminex column (BioRad) was used with injection volumes of 10 μ L. The run time was 60 min at a flow rate of 0.3 mL/min and at 65 °C. Calibration was conducted with concentration standards between 0.1 and 5 mM. The standards were: 2,3-furandiols, methylglyoxal, and ethylene glycol, in 0.5 M $KHCO_3$, detected using the RID. Acetic acid and formic acid standards were prepared at concentrations of 0.01–5 mM and detected by UV at 210 nm. Product assignment was confirmed by ¹H NMR, as described in detail in the SI (Figure S17).

X-ray Photoelectron Spectroscopy (XPS). The surface of Fe_2P before and after reaction was analyzed by a Thermo K-Alpha XPS spectrometer. The chamber was evacuated to 5×10^{-9} Torr base pressure. The spectra were collected with a flood gun for charge compensation, and an X-ray beam of 400 μ m was used.

Computational Methods. DFT and GC-DFT calculations with periodic boundary conditions were performed using the JDFTx code.⁶⁷ The generalized gradient approximation (GGA) was employed to compute the exchange-correlation energy using the Perdew–Burke–Ernzerhof (PBE)⁶⁸ functional combined with the SG15 Optimized Norm-Conserving Vanderbilt (ONCV) pseudopotentials to implicitly model the core electrons of the system.^{69,70} The algorithm implemented by JDFTx variationally minimized the grand free energy at a set constant potential by varying the electron count within the unit cell, which were balanced by fluid bound charges (see SI for a more detailed description of GC-DFT).

The bare Fe_2P (0001) surface was modeled using a four-layer supercell consisting of alternating layers with Fe_3P_2 and Fe_3P stoichiometries, making a total of 48 Fe and 24 P atoms in the supercell to reduce undesired adsorbate–adsorbate interactions between periodic images in neighboring cells. The lattice parameters were optimized for the clean surface and fixed for all adsorbate and reconstruction calculations. A $2 \times 2 \times 1$ Monkhorst–Pack grid was used for k-point sampling of the Brillouin zone. Geometries were optimized with a 20 hartree plane wave energy cutoff energy. The electronic energies were converged to within 1×10^{-8} hartree. Relaxation of the magnetic states resulted in the ferrimagnetic state, as expected for Fe_2P . The charge-asymmetry corrected, local-response, nonlocal-cavity solvation model (CANDLE) was implemented to account for solvation effects on molecules and surfaces.⁷¹ An electrolyte consisting of 0.5 M Na^+ and 0.5 M F^- was embedded into the solvent model.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.1c03428>.

Characterization data of catalyst by PXRD and SEM, GC-DFT reference states and reconstruction, faradaic efficiency, current density, pDOS plots, catalyst stability assessment through ICP and XPS, ECSA, EIS, NMR, adsorption energies, and details of the calculations (PDF)

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Notes

The authors declare no competing financial interest.

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